

THE SOLUTION –
SUBSYSTEM MOATS

SWARM SEARCH
EFFICIENCY

AI Swarm Drones Find Survivors 3–4x Faster Than Traditional SAR

- 78-85% survivor detection on first pass
- 10-18 min per sq mile speed
- Mesh networking works in GPS-denied zones
- 24/7 autonomous operation for 11+ days
- False positive rate down to 12%
- Cost per deployment: \$12K vs \$15K

PROJECT SUTRA SUBSYSTEM SOLUTIONAL
MOATS

3 CORE ENGINEERING
MOATS

SUBSYSTEM
A MOAT 50Hz
OFFBOARD

PX4 + VIO EKF2 Autopilot

Decentralized 3D ORCA anti-collision and offboard setpoint streaming without ground

SUBSYSTEM
B MOAT -5dB
RESILIENT

Deep JSCC Neural Video

Zero-cliff neural semantic video encoding that degrades gracefully under extreme radio jamming and ravine occlusion.

SUBSYSTEM
C/D MOAT 3.59cm
ACCURACY

3D DEM Terrain Raycasting

Sub-15ms edge YOLOv8-Pose with WGS84 elevation raycasting, eliminating 15-30m flat-Earth

control tower
dependency.

geolocation
errors.